电路分析基础总复习

一、电路分析的基本参量:

电压$u(t)$: 单位V，当 $u(t)=U_{\mathrm{m}}\cos(\omega t+\varphi_u)$ 时，

$$\dot{U}_{\mathrm{m}}=U_{\mathrm{m}}\mathrm{e}^{\mathrm{j}\varphi_u}=U_{\mathrm{m}}\angle\varphi_u \quad --\text{电压振幅相量}$$

$$\dot{U}=U\mathrm{e}^{\mathrm{j}\varphi_u}=U\angle\varphi_u \quad --\text{电压有效值相量}$$

电流$i(t)$: 单位A，当 $i(t)=I_{\mathrm{m}}\cos(\omega t+\varphi_i)$ 时，

$$\dot{I}_{\mathrm{m}}=I_{\mathrm{m}}\mathrm{e}^{\mathrm{j}\varphi_i}=I_{\mathrm{m}}\angle\varphi_i \quad --\text{电流振幅相量}$$

$$\dot{I}=I\mathrm{e}^{\mathrm{j}\varphi_i}=I\angle\varphi_i \quad --\text{电流有效值相量}$$

关联参考方向：电流参考方向与电压参考极性方向一致。

功率：单位**W**

$p(t) = u(t)i(t)$ 关联参考方向

$p(t) = -u(t)i(t)$ 非关联参考方向

$p > 0$, 吸收(消耗)功率； $p < 0$, 提供(产生)功率

二、电路的基本元件:

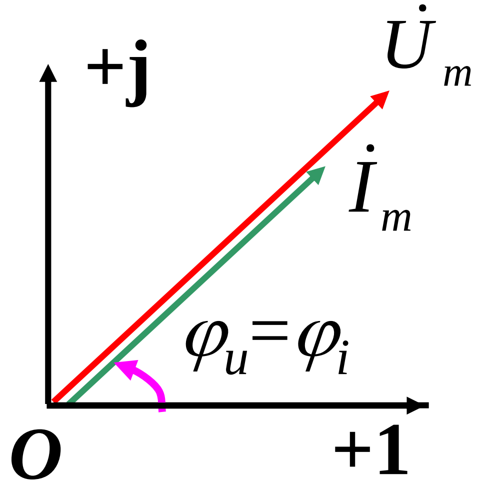
1. 电阻：单位Ω；　　**电导**：单位S，$G=\dfrac{1}{R}$

$$u=Ri$$

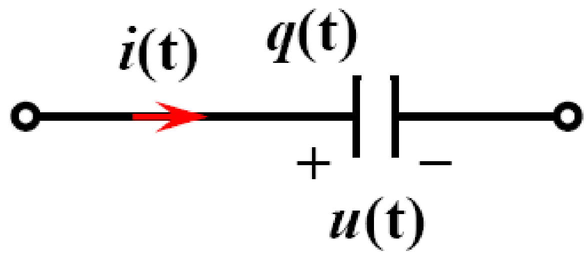
$$Z_R=R \qquad Y_R=\frac{1}{R}=G$$

$$\dot{U}_m=R\dot{I}_m$$

$$\begin{cases} U_m=RI_m \\ \angle\varphi_u=\angle\varphi_i \end{cases}$$



2. **电容**：单位F，$C=\dfrac{q(t)}{u(t)}$



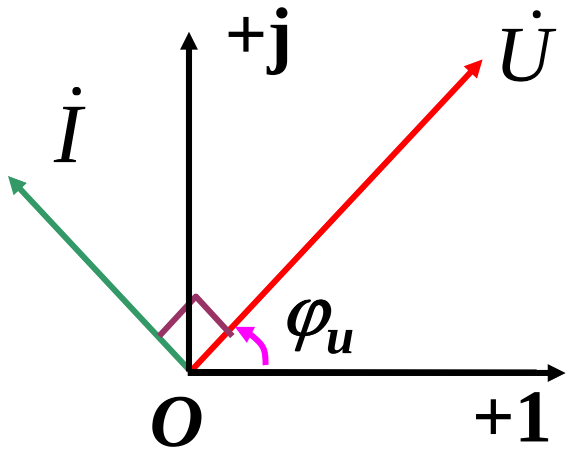
$$i(t)=C\frac{du_C}{dt} \qquad u_C(t)=u_C(t_0)+\frac{1}{C}\int_{t_0}^{t} i(\xi)d\xi$$

$$W_C(t)=\frac{1}{2}Cu_C^2(t) \qquad u_C(t_{0-})=u_C(t_{0+})$$

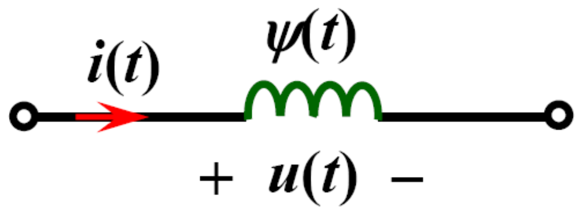
$$Z_C=-\mathrm{j}\frac{1}{\omega C} \qquad X_C=-\frac{1}{\omega C}$$

$$Y_C=\mathrm{j}\omega C \qquad B_C=\omega C$$

$$\dot{I}_m=\mathrm{j}\omega C\dot{U}_m \qquad \begin{cases} I_m=\omega CU_m \\ \angle\varphi_i=\angle\varphi_u+90^{\circ} \end{cases}$$



3. 电感：单位H，$L = \dfrac{\psi(t)}{i(t)}$



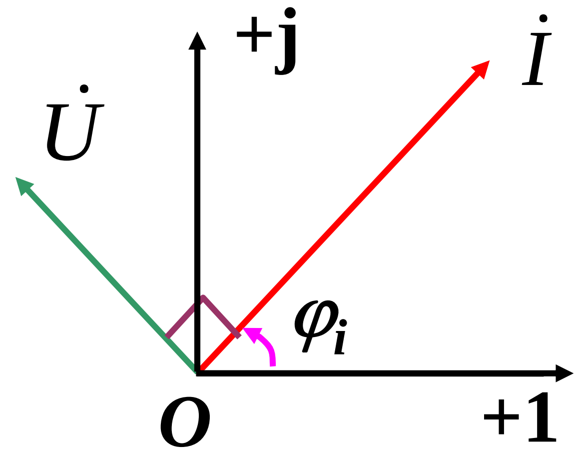
$$u(t) = L\frac{di_L}{dt} \qquad i_L(t) = i_L(t_0) + \frac{1}{L}\int_{t_0}^{t} u(\xi)d\xi$$

$$W_L(t) = \frac{1}{2}Li_L^2(t) \qquad i_L(t_{0-}) = i_L(t_{0+})$$

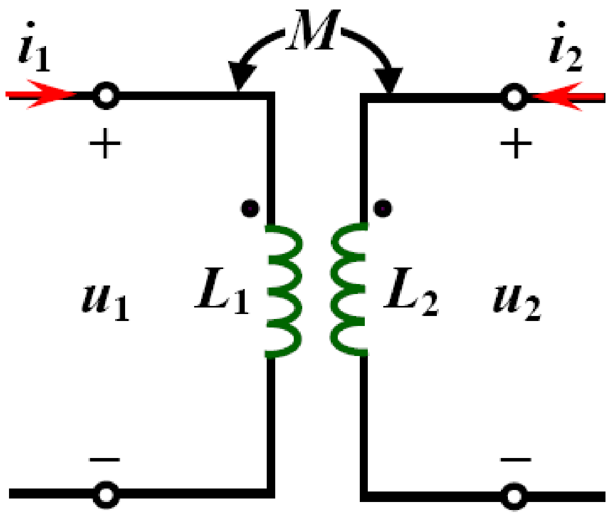
$$Z_L = \mathrm{j}\omega L \qquad X_L = \omega L$$

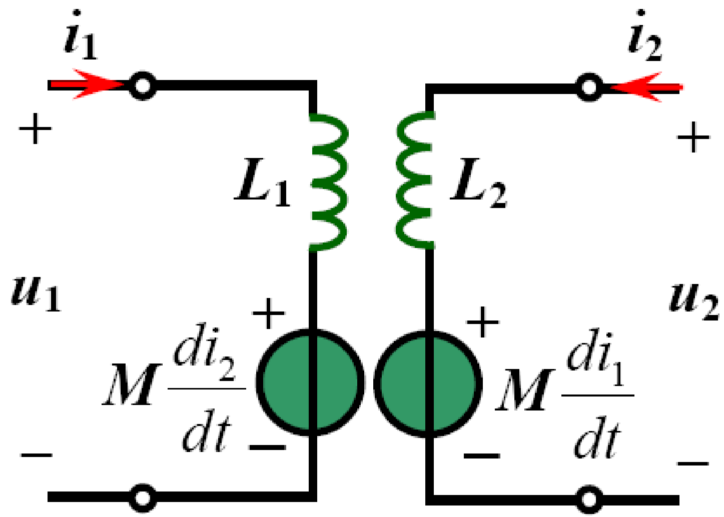
$$Y_L = -\mathrm{j}\frac{1}{\omega L} \qquad B_L = -\frac{1}{\omega L}$$

$$\dot{U}_m = \mathrm{j}\omega L\dot{I}_m \qquad \begin{cases} U_m = \omega L I_m \\ \angle\varphi_u = \angle\varphi_i + 90^\circ \end{cases}$$



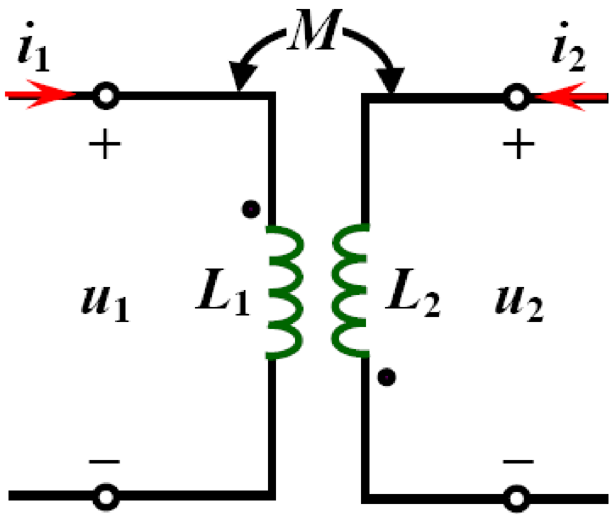
4. 耦合电感

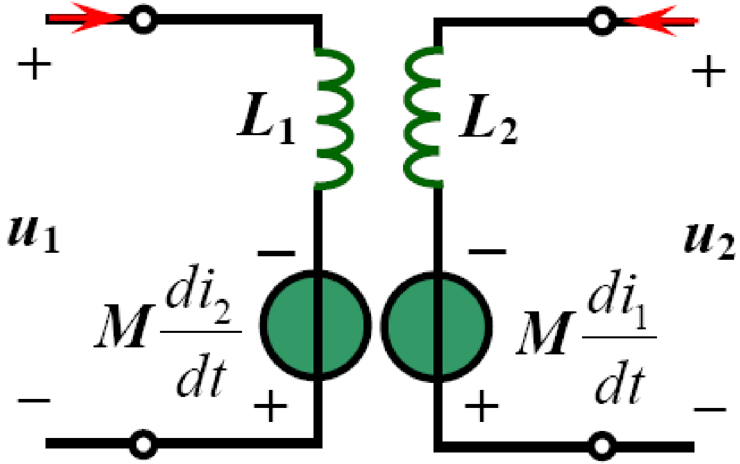




$$\begin{cases} u_1 = L_1 \dfrac{di_1}{dt} + M \dfrac{di_2}{dt} \\ u_2 = L_2 \dfrac{di_2}{dt} + M \dfrac{di_1}{dt} \end{cases}$$

$$\begin{cases} \dot{U}_1 = \mathrm{j}\omega L_1 \dot{I}_1 + \mathrm{j}\omega M \dot{I}_2 \\ \dot{U}_2 = \mathrm{j}\omega L_2 \dot{I}_2 + \mathrm{j}\omega M \dot{I}_1 \end{cases}$$

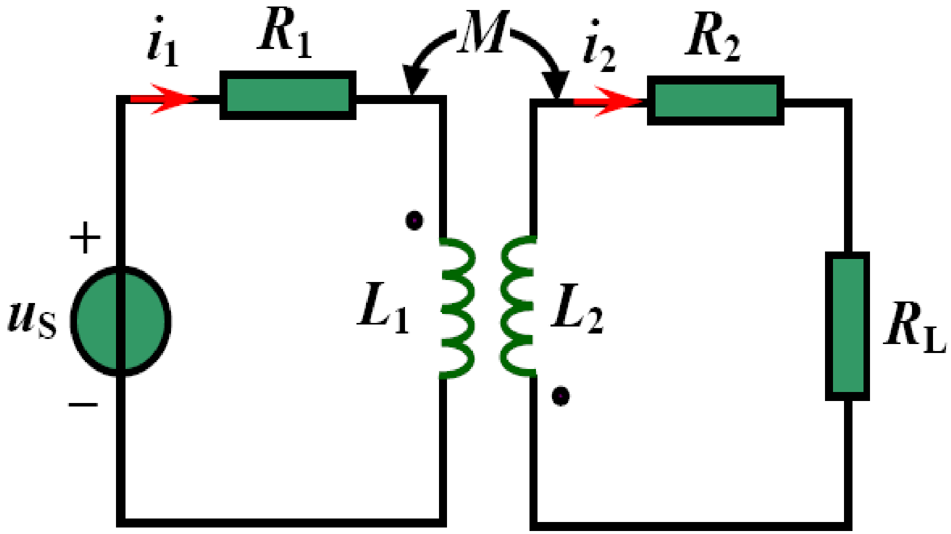




$$\begin{cases} u_1 = L_1\dfrac{di_1}{dt} - M\dfrac{di_2}{dt} \\ u_2 = L_2\dfrac{di_2}{dt} - M\dfrac{di_1}{dt} \end{cases}$$

$$\begin{cases} \dot{U}_1 = \mathrm{j}\omega L_1\dot{I}_1 - \mathrm{j}\omega M\dot{I}_2 \\ \dot{U}_2 = \mathrm{j}\omega L_2\dot{I}_2 - \mathrm{j}\omega M\dot{I}_1 \end{cases}$$

反映阻抗: $Z_{ref} = Z_{反} = \dfrac{\omega^2 M^2}{Z_{22}}$



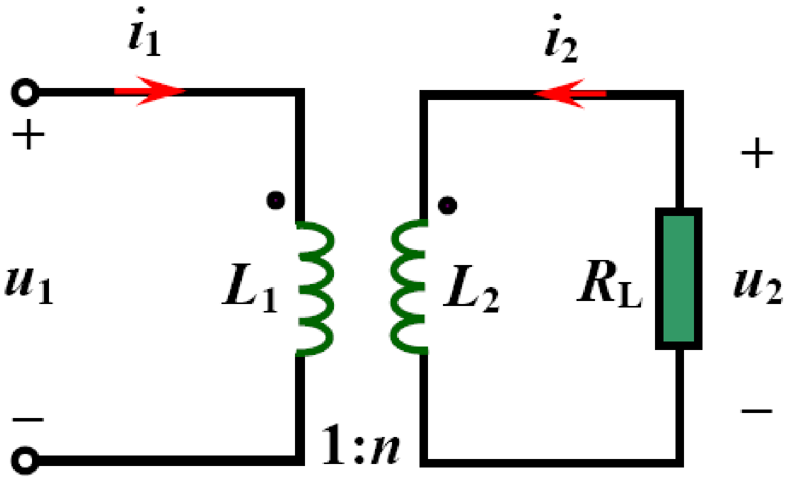
输入阻抗:

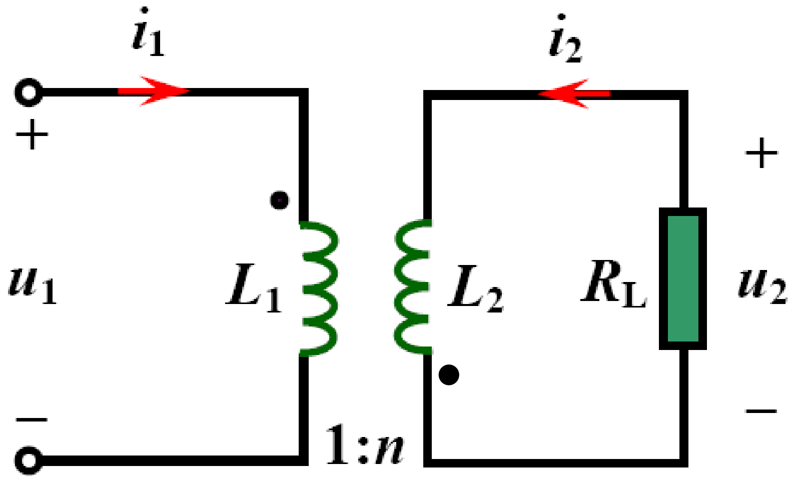
$$Z_i = Z_{11} + \frac{\omega^2 M^2}{Z_{22}}$$

二次电流:

$$\dot{I}_2 = \frac{-\mathrm{j}\omega M \dot{I}_1}{Z_{22}}$$

5. 理想变压器





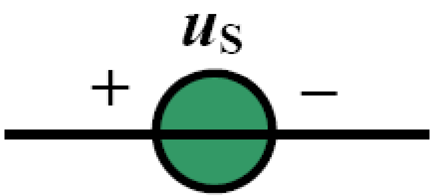
$$\begin{cases} u_2 = nu_1 \\ i_2 = -\dfrac{1}{n} i_1 \end{cases}$$

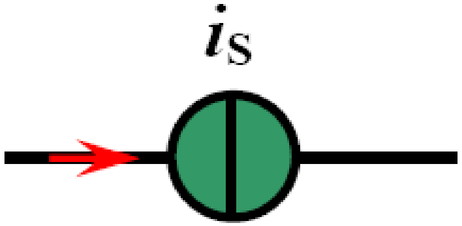
$$R_i = \frac{1}{n^2} R_L$$

$$\begin{cases} u_2 = -nu_1 \\ i_2 = \dfrac{1}{n} i_1 \end{cases}$$

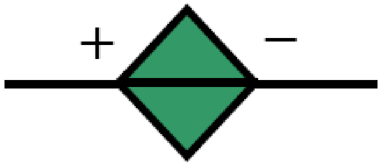
$$R_i = \frac{1}{n^2} R_L$$

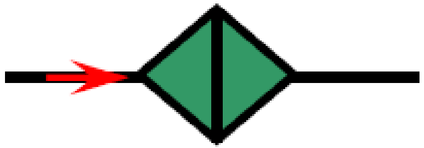
6. 独立源





7. 受控源





CCVS **CCCS**

VCVS **VCCS**

三、基本电路:

时域模型、相量模型（复数域模型）

电阻电路、动态电路（一阶电路、二阶电路）

四、元件及电路的约束方程:

欧姆定律:　$u = Ri$　　$\dot{U}_m = Z\dot{I}_m$

基尔霍夫定律:　**KCL:**　$\sum_{k=1}^{K} i_k = 0$　　$\sum_{k=1}^{K} \dot{I}_{km} = 0$

KVL:　$\sum_{k=1}^{K} u_k = 0$　　$\sum_{k=1}^{K} \dot{U}_{km} = 0$

五、电路的分析方法：

1. 应用基尔霍夫定律分析；
2. 支路分析；
3. 网孔电流分析；
4. 节点电压分析；
5. 利用线性电路的性质分析（互易性、叠加性）；
6. 等效变换法分析：电源等效变换、戴维南（诺顿）等效电路、元件等效置换、T形（Π形）网络等效变换、正弦量变换为相量；
7. 三要素法分析；
8. 分解法分析。

六、线性网络的重要定理:

1. 叠加定理;　　2. 互易定理;

3. 置换定理;　　4. 戴维南（诺顿）定理;

5. 最大功率传递定理。

七、电路的响应:

零输入响应、零状态响应、完全响应

瞬态响应、稳态响应

阶跃响应、冲激响应